

MarketMind AI – Intelligent Market Analysis & WhatsApp Alert Agent

Abstract

MarketMind AI is a financial intelligence agent designed to help retail investors and traders make faster and smarter market decisions using real-time institutional activity, global market news, and market insights.

The platform continuously collects and analyzes stock market data including daily buy/sell activity, bulk and block deals, promoter transactions, institutional inflows/outflows (FII/DII), and global financial news from major economies such as the United States, Europe, and China.

The system identifies unusual market movements, sector-wise momentum, and potential investment signals. The platform delivers insights directly through WhatsApp notifications, enabling users to stay updated in real time.

Unlike traditional dashboards, MarketMind AI acts as an autonomous financial assistant capable of monitoring markets continuously, explaining why a stock or sector is moving, and proactively alerting users about critical events such as abnormal institutional buying, sudden volume spikes, or major global economic announcements.

Key Features

- Stock Intelligence Engine
 - Global Market News Analyzer
 - Sector Heatmap & Trend Analysis
 - Unusual Activity Detection System
 - Explainable Recommendation Engine
 - WhatsApp Alert Agent
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Stock Intelligence Engine

- Daily buy/sell analysis
 - Bulk and block deal tracking
 - Promoter buy/sell monitoring
 - FII/DII activity analysis
 - Signal generation: Buy / Hold / Reduce Exposure
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Global Market News Analyzer

- Fetches news from US, Europe, and China markets
 - News classification: Bullish / Bearish / Neutral
 - Detects impacted sectors such as Banking, IT, Energy, Pharma, and Semiconductor
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Sector Heatmap & Trend Analysis

- Weekly sector performance tracking
 - Global market trend graphs
 - Risk-on / Risk-off sentiment analysis
 - Comparative market movement visualization
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Unusual Activity Detection System

Detects:

- Sudden volume spikes
 - Heavy institutional accumulation
 - Abnormal promoter activity
 - High-impact bulk deals
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Explainable Recommendation Engine

The system explains the reasoning behind each signal.

Example:

BUY Signal for Reliance

Reason:

- Strong FII buying
 - Positive refining margin news
 - Bullish energy sector momentum
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WhatsApp Alert Agent

Automated delivery of:

- Top market movers
 - Institutional activity summaries
 - Global market alerts
 - Sector trend reports
 - Charts and graphs
 - Event-driven notifications
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Technology Stack

Backend:

- Java Spring Boot
- REST APIs
- Spring Scheduler

- Spring Data JPA

Frontend:

- React.js
- Chart.js / Recharts
- Material UI

Database:

- PostgreSQL / MongoDB

Notifications:

- Twilio WhatsApp API
- WhatsApp Business API

Deployment:

- Docker
 - AWS / Render / Railway
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Innovation & Impact

MarketMind AI transforms raw financial data into explainable and actionable market insights. By combining institutional data, news analysis, and automated messaging, the platform reduces information overload and enables faster investment decisions.

Future Enhancements

- Portfolio risk analysis
 - Swing trade detection
 - Voice-enabled market assistant
 - Personalized investment strategies
 - Telegram & Email integrations
 - Real-time chatbot for stock queries
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Expected Outcome

A fully functional market intelligence agent capable of:

- Monitoring stock markets continuously
- Detecting high-impact market activity
- Explaining market movements intelligently
- Delivering actionable insights instantly through WhatsApp